KAIZEN MATHS

Lesson starter

Bearings

Bearings and Trigonometry

Name _____

SA1



A ship travels 15 km from A on a bearing of 117°. Find its east-west and north-south displacement.

GEO-5B87-L3 · SA

Show your working

KAIZEN MATHS

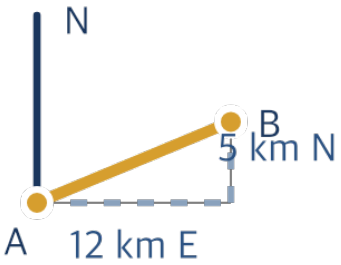
Lesson starter

Bearings

Bearings and Trigonometry

Name _____

SB1



From A, B is 12 km east and 5 km north. Find the distance and bearing of B from A.

GEO-5B87-L3 · SB

Show your working

KAIZEN MATHS

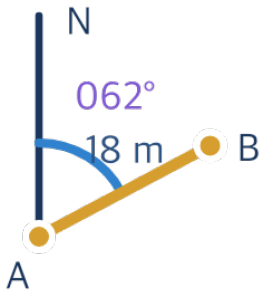
Lesson starter

Bearings

Bearings and Trigonometry

Name _____

SC1



A ship travels 18 m from A on a bearing of 062°. Find its east-west and north-south displacement.

GEO-5B87-L3 · SC

Show your working

KAIZEN MATHS

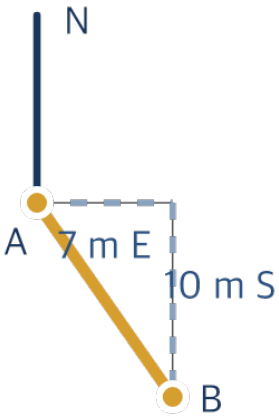
Lesson starter

Bearings

Bearings and Trigonometry

Name _____

SD1



From A, B is 7 m east and 10 m south. Find the distance and bearing of B from A.

GEO-5B87-L3 · SD

Show your working

KAIZEN MATHS

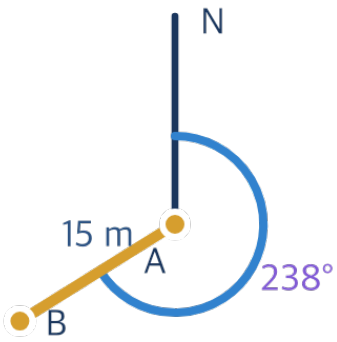
Exit ticket

Bearings

Bearings and Trigonometry

Name _____

EA1



A ship travels 15 m from A on a bearing of 238°. Find its east-west and north-south displacement.

GEO-5B87-L3 · EA

Show your working

KAIZEN MATHS

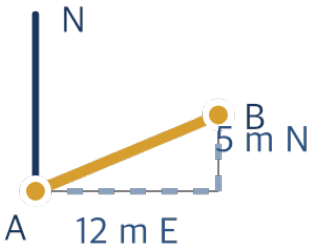
Exit ticket

Bearings

Bearings and Trigonometry

Name _____

EB1



From A, B is 12 m east and 5 m north. Find the distance and bearing of B from A.

GEO-5B87-L3 · EB

Show your working

KAIZEN MATHS

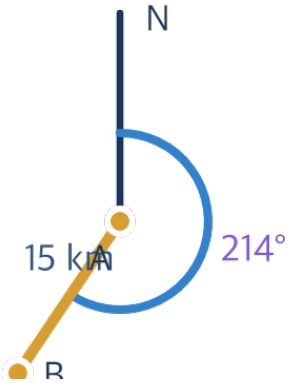
Exit ticket

Bearings

Bearings and Trigonometry

Name _____

EC1



A ship travels 15 km from A on a bearing of 214°. Find its east-west and north-south displacement.

GEO-5B87-L3 · EC

Show your working

KAIZEN MATHS

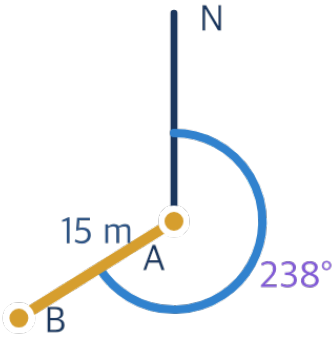
Exit ticket

Bearings

Bearings and Trigonometry

Name _____

EA1



A ship travels 15 m from A on a bearing of 238°. Find its east-west and north-south displacement.

GEO-5B87-L3 · EA

Show your working

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



A ship travels 15 km from A on a bearing of 117°. Find its east-west and north-south displacement.

Answer:

13.4 km east, 6.8 km south

Working:

Step 1: Draw a right triangle using the journey line as the hypotenuse.

Step 2: The east-west component is opposite the bearing angle, so use d sin(θ).

East-west component = 15 sin(117°) ≈ 13.4 km.

Step 3: The north-south component is adjacent to the bearing angle, so use d cos(θ).

North-south component = 15 cos(117°) ≈ 6.8 km.

Step 4: Use the quadrant of the bearing to decide the direction words.

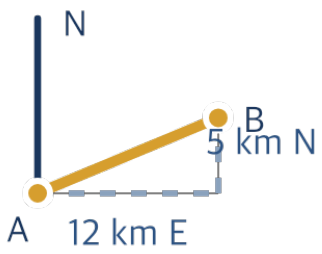
Displacement = 13.4 km east, 6.8 km south.

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



From A, B is 12 km east and 5 km north. Find the distance and bearing of B from A.

Answer:

13.0 km, bearing 067°

Working:

Step 1: The east-west and north-south movements form a right triangle.

Step 2: Use Pythagoras to find the distance AB.

$AB = \sqrt{(12^2 + 5^2)} \approx 13.0\ \text{km}.$

Step 3: Use tangent to find the acute angle from the nearest north-south line.

$\tan(\theta) = 12 \div 5$

$\theta \approx 67.4°$

Step 4: Decide the bearing using the quadrant. NE quadrant, so the bearing is 67.4° from north.

Bearing ≈ 067°.

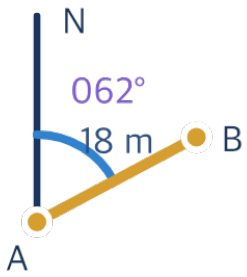
Distance and bearing = 13.0 km, bearing 067°.

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



A ship travels 18 m from A on a bearing of 062°. Find its east-west and north-south displacement.

Answer:

15.9 m east, 8.5 m north

Working:

Step 1: Draw a right triangle using the journey line as the hypotenuse.

Step 2: The east-west component is opposite the bearing angle, so use d sin(θ).

East-west component = 18 sin(62°) ≈ 15.9 m.

Step 3: The north-south component is adjacent to the bearing angle, so use d cos(θ).

North-south component = 18 cos(62°) ≈ 8.5 m.

Step 4: Use the quadrant of the bearing to decide the direction words.

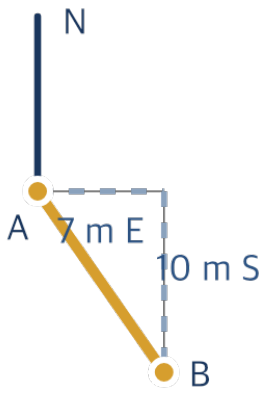
Displacement = 15.9 m east, 8.5 m north.

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



From A, B is 7 m east and 10 m south. Find the distance and bearing of B from A.

Answer:

12.2 m, bearing 145°

Working:

Step 1: The east-west and north-south movements form a right triangle.

Step 2: Use Pythagoras to find the distance AB.

$AB = \sqrt{(7^2 + 10^2)} \approx 12.2$ m.

Step 3: Use tangent to find the acute angle from the nearest north-south line.

$\tan(\theta) = 7 \div 10$

$\theta \approx 35.0°$

Step 4: Decide the bearing using the quadrant. SE quadrant, so the bearing is 180° - 35.0°.

Bearing ≈ 145°.

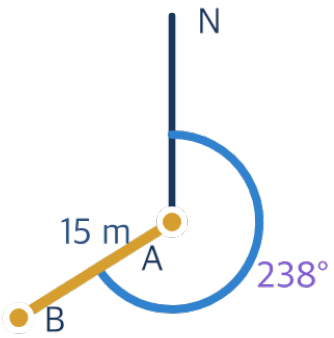
Distance and bearing = 12.2 m, bearing 145°.

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



A ship travels 15 m from A on a bearing of 238°. Find its east-west and north-south displacement.

Answer:

12.7 m west, 7.9 m south

Working:

Step 1: Draw a right triangle using the journey line as the hypotenuse.

Step 2: The east-west component is opposite the bearing angle, so use d sin(θ).

East-west component = 15 sin(238°) ≈ 12.7 m.

Step 3: The north-south component is adjacent to the bearing angle, so use d cos(θ).

North-south component = 15 cos(238°) ≈ 7.9 m.

Step 4: Use the quadrant of the bearing to decide the direction words.

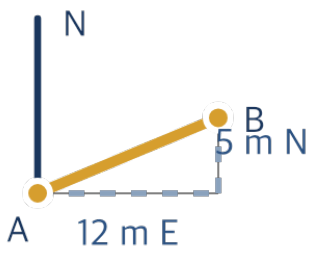
Displacement = 12.7 m west, 7.9 m south.

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



From A, B is 12 m east and 5 m north. Find the distance and bearing of B from A.

Answer:

13.0 m, bearing 067°

Working:

Step 1: The east-west and north-south movements form a right triangle.

Step 2: Use Pythagoras to find the distance AB.

$AB = \sqrt{(12^2 + 5^2)} \approx 13.0$ m.

Step 3: Use tangent to find the acute angle from the nearest north-south line.

$\tan(\theta) = 12 \div 5$

$\theta \approx 67.4°$

Step 4: Decide the bearing using the quadrant. NE quadrant, so the bearing is 67.4° from north.

Bearing ≈ 067°.

Distance and bearing = 13.0 m, bearing 067°.

Teacher answers & working

Bearings · Bearings and Trigonometry

GEO-5B87-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



A ship travels 15 km from A on a bearing of 214°. Find its east-west and north-south displacement.

Answer:

8.4 km west, 12.4 km south

Working:

Step 1: Draw a right triangle using the journey line as the hypotenuse.

Step 2: The east-west component is opposite the bearing angle, so use d sin(θ).

East-west component = 15 sin(214°) ≈ 8.4 km.

Step 3: The north-south component is adjacent to the bearing angle, so use d cos(θ).

North-south component = 15 cos(214°) ≈ 12.4 km.

Step 4: Use the quadrant of the bearing to decide the direction words.

Displacement = 8.4 km west, 12.4 km south.